A Model for Optimization of Food Bank Allocation and Routing

An Evaluation

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Overview

Introduction

Motivation

- **Definition:** Food insecurity is the state of having reliable access to a sufficient quantity of affordable, nutritious food.
- **Role of Food Banks:** Organizations like the Food Bank of the Southern Tier (FBST) look to support these needs by redistributing food to vulnerable populations.
- **Logistical Complexity:** Despite these efforts, the prevalence of food insecurity has continued to increase according to the USDA, due to:
  - Limited resources
  - Distance in partner sites
  - Seasonal factors such as winter weather

Food insecurity - U.S. Department of Agriculture

In the U.S., 13.5% (18 million) of households were food insecure at some time during 2023.

Objective

- **Question:** How can food banks distribute resources efficiently while maintaining fairness?
- **Objective:** Develop an optimization model that:
    - Accounts for transportation constraints (distances, time, etc.)
    - Considers weather in demand.
    - Balances efficiency (total utilization) with fairness between sites.
- **Approach:**
    - Formulate a routing model and compare it against historical (2019) data.
    - Incorporate winter weather constraints.
    - Propose a Fisher market model to handle stochastic resource demands.
- **Our contribution:**
    - Provide insights to inform policy and operational decisions.
    - Emphasize tradeoffs between serving the most people versus providing coverage.

Methodology

Demand Analysis

- Simulated demand using random normal distribution with our mean and standard deviation from MFP Demand Data (2019). This includes data related to:
  - Density, Household income, Food stamps, Poverty, Number of children, Total population
- Scaled our demand data and used Principal Component Analysis
  - PCA synthesizes the information from large datasets into a smaller set of uncorrelated variables known as "principal components."
  - Use case: to remove correlation between features in MFP Demand Data.
- Created three linear combinations from our processed data – explaining $> 90\%$ of the variance in the demand
- Regression on the original data has a very large condition number (order of 10^7), hinting at near-multicollinearity.
- Comparing this to the regression results from PCA, all components are significant.

Sample Household Generation

- In order to better estimate and sample characteristics of the various regions, sample households were created
- Samples were drawn using an acceptance-rejection algorithm
  - Sample were drawn uniformly within each PUMA according to its population-weighted density
- Characteristics like household size, income, poverty status, access to a car etc. were generated according to parameterized distributions fit on American Community Survey (ACS) data
- Households were then typed using the previous PCA analysis
- Each food bank was then assigned (non mutually-exclusively) all households within 25 miles of them to estimate the incidence rates of types and transportation access

Sample Household Generation

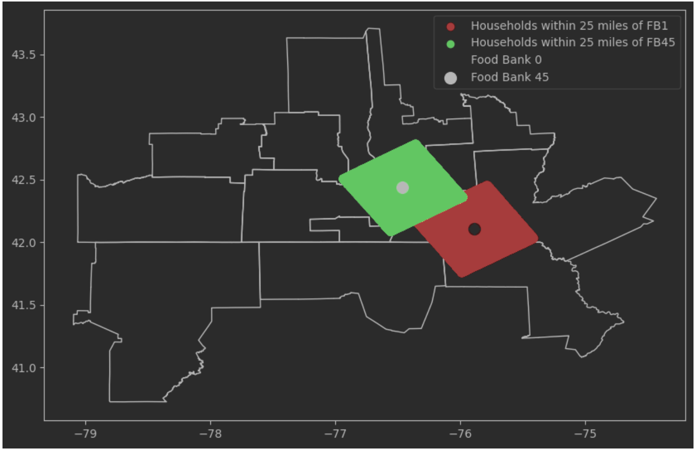

Figure: Generated Households within 25 miles of Food Bank 1 (red) and Food Bank 45 (green)

Standard Routing Model

- Optimization program over all 70 food banks (i) for the whole year
  - One year $=$ 260 days (j) (52 weeks $\times$ 5 business days)
- Each truck/set of trucks makes three visits (k) per day with no repeated sites in a day
  - Average number of visits per day was $\sim$2.8 in 2019
- Maximum travel distance (d) for a truck in a given day was 200 miles (m), excluding travel to and from resupply
  - Unreasonable for a truck to be traveling vast distances in one day while leaving time for distribution
  - Distance was calculated as the Manhattan distance between each food bank using the Conus Albers projection system in meters
- A visit to a site at any point in the year grants utility (u) equal to the average demand in 2019
  - For now we assume utility per consumer is the same and deterministic
- Optimal solution benchmarked against the 2019 allocation as measured by average number of visits to each site

Standard Routing Model

- Objective: Maximize utility of 'effective' visits

$$\max_{x_{i,j,k}} \sum_i u_i \sqrt{cV_i} \qquad V_i = \sum_{j,k} x_{i,j,k} \forall i$$

- Effective visits are $\sqrt{cV_i}$, representing decreasing marginal returns to visits to the same site
- $\sqrt{cV_i} > V_i$ for $V_i < c$ and $\sqrt{cV_i} = V_i$ for $V_i = c$
  - This gives c a natural interpretation of the number of visits needed to 'saturate demand'
  - Lower c means more dispersed visits
  - $c = 5$ in this model to penalize concentration at high demand sites while accounting for true average number of visits

Standard Routing Model

- Only one of each site can be visited per day

$$\sum_{k} x_{i,j,k} \leq 1, \forall i, j$$

- Only one site per spot

$$\sum_{i} x_{i,j,k} \leq 1, \forall j, k$$

- The maximum distance traveled per day is less than 200 miles

$$\sum_{i} \sum_{l, l>i} \sum_{k \in \{0,1\}} d_{i,l} y_{i,l,j,k} \leq m, \forall j$$

where y is an indicator of whether sites i and l were visited in succession on a given day

Evaluating the Standard Model

- Four metrics were constructed to evaluate the efficiency and fairness characteristics of the solution

Utilitarianism (Efficiency)

Let V_i be the total (unpenalized) visits to site i, and u_i the utility for doing so. Then the total utility is $\sum_i V_i u_i$

- The rationale for this is that utilitarianism is Pareto Efficient where utility is quasi-linear with monetary transfers
- Since we can express the food allocation in dollar terms it's a reasonable assumption here

Evaluating the Standard Model

Efficiency Gap (Fairness)

Let V_i^{2019} be the number of visits in 2019. Then $\sum_i \min(0, V_i u_i - V_i^{2019} u_i)$ is the gross loss suffered by sites that receive less visits in an allocation compared to 2019

- The efficiency gap is the total amount of utility needed to be reallocated to the sites that lose utility in order for the current solution to make no site worse off
- This metric is closely related to Pareto Efficiency as it's the utility value of the monetary transfers needed to construct a Pareto improvement

Evaluating the Standard Model

Coefficient of Variation (Fairness)

Let μ, σ be the sample mean and sample standard deviation of utilities in an allocation. Then the coefficient of variation is $\frac{\mu}{\sigma}$

- This is a measure of the relative spread of the utilities
- CV, while not commonly used as a measure of inequality, does satisfy all the requisite qualities to be one
- CV $= 1$ is typically thought of to be moderate variation

Gini Coefficent (Fairness)

Let μ be the sample mean of the utilities. Then $\frac{\sum_{i=1}^{n} \sum_{j=1}^{n} |u_i - u_j|}{2n^2\mu}$ is the Gini coefficient

- Gini coefficient is a widely used measure of economic inequality
- 1 is absolute inequality (one has all the wealth) and 0 is absolute equality (all wealth is equal)

Incorporating Winter Weather

- A sub-problem over winter days was solved to account for the impact of winter
  - Winter in this model constitutes roughly the months of January, February, November and December
  - 64 days total
- To account for the impact of winter on demand, the average demand per visit was scaled by the fraction of households within 25 miles that had access to a car
  - While other methods of transportation exist, the rates for public transportation, biking etc. were all negligible
- Maximum number of sites was constrained to be 20 total
  - Model is forced to pick the best subset of sites to service

Incorporating Winter Weather

- New set of constraints implementing subset selection:

$$x_{i,j,k} \leq Mz_i, \forall i, j, k$$

$$\sum_i z_i \leq 20$$

  where M is a large constant implementing the "big-M" or logic here
- This model was benchmarked against both the unconstrained winter allocation as well as the 2019 allocation
  - Measured on the same four metrics defined earlier
  - The 2019 allocation here is the fraction of the year that winter is times the average number of visits in 2019

Stochastic Allocation

- The final model is one that allocates a set of four resources (k) over a set of fixed sites (t)
  - The fixed route was the 12 sites with the highest average demand
  - The four resources were derived from USDA eating guidance and are: dairy, fruits and vegetables (greens), grains and protein
  - The total capacity of 180,000 lbs was chosen in order to make the problem feasible
- Additionally, a set of household types θ were constructed based on the results of the PCA analysis earlier
- Using the household sampling procedure earlier, the fraction of the population that is each type in range of each food bank was constructed
  - The fraction is in fact a normal distribution fit using the mean and standard deviation of each type multiplied by the average demand per visit from 2019
- Each type has different weights α for each resource, as well as a scaling factor A
  - α was derived using the previous good consumption analysis
  - A was chosen based on the characteristics of each type (household income, number of children etc.)

Stochastic Allocation

- The problem objective is:

$$\max \sum_{\theta} A_{\theta} \sum_{t} N_{t,\theta} \log(u_{\theta}(x_{t,\theta,k}))$$

- With Cobb-Douglas utility:

$$u_{\theta} = \prod_{k} x_{t,\theta,k}^{\alpha_{\theta,k}}$$

- And capacity constraint:

$$\sum_{\theta,t} N_{t,\theta} x_{t,\theta,k} \leq c_k, \forall k$$

Stochastic Allocation

- However, the number of families is not deterministic here and so a certainty equivalence procedure was implemented
- The procedure is as follows:
  - Move to site t and realize arrivals $N_{t,\theta}$
  - All future sites have arrivals set to their expectation $\mathbb{E}[N_{t',\theta}]$ for $t' > t$
  - An allocation $x_{t,\theta,k}$ is generated
  - The total resources allocated at that step $\sum_{\theta} N_{t,\theta} x_{t,\theta,k}$ is subtracted from the capacity c_k
  - The process is repeated at the next site with the updated capacity until all sites have been allocated to
- We denote the above as the myopic solution
- We benchmark the myopic solution against the hindsight solution, which is a one-shot procedure that uses the realized demands and solves in advance

Evaluating Stochastic Allocation

Hindsight Gap (Efficiency)

Let u_{myopic}, $u_{hindsight}$ be the total utility of the myopic and hindsight solutions, respectively. Then the hindsight gap is $\frac{u_{hindsight} - u_{myopic}}{u_{hindsight}}$

- The hindsight gap measures how much better the hindsight solution is compared to the myopic solution

Maximum Envy (Fairness)

Let $u_{t,\theta} = \sum_k u_{t,\theta,k}$ be the total utility at each site t for each type θ. Then the maximum envy for each type is $E_\theta = \max(u_{t',\theta} - u_{t,\theta}, \forall t, t' \neq t)$. The total maximum envy is $\sum_\theta E_\theta$

- Maximum envy measures how much a given household type could improve by being at a different site
- The aggregate metric is simply the total amount of 'envy' present

Results and Discussion

Demand Analysis – Tabular Explanation

Variable	Coef.	Std. Err.	t	p-value	[0.025	0.975]
const	-171.7447	63.886	-2.688	0.007	-297.172	-46.318
NUMPREC	127.8586	64.644	1.978	0.048	0.944	254.773
DENSITY	-0.0507	0.014	-3.542	0.000	-0.079	-0.023
HHINCOME	-0.0023	0.002	-1.188	0.235	-0.006	0.001
FOODSTMP	-5.4136	11.994	-0.451	0.652	-28.961	18.134
POVERTY	-1.6749	0.959	-1.747	0.081	-3.557	0.207
NCHILD	248.3854	71.665	3.466	0.001	107.688	389.083
PUMA_pop	0.0041	0.001	3.401	0.001	0.002	0.007

Variable	Coef.	Std. Err.	z	p-value	[0.025	0.975]
const	143.2850	3.479	41.192	0.000	136.467	150.103
PC1	-0.0017	0.000	-4.057	0.000	-0.002	-0.001
PC2	-180.8965	31.031	-5.830	0.000	-241.716	-120.077
PC3	-626.5524	176.689	-3.546	0.000	-972.856	-280.249

Demand Analysis – Graphical Representation

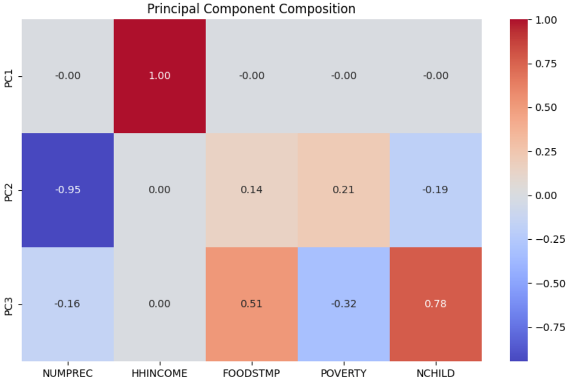

Figure: Heatmap describing the three principal component features

Standard Routing Model

- As expected, the optimal solution has a substantially higher total utility, thus performing much better on the efficiency metric
- However, it does this by visiting high utility sites more often, leading to a much more unequal allocation
  - The Gini Coefficient almost doubling is particularly important, as even small changes of 0.01 on this measure
- Looking at the efficiency gap, it is possible to construct transfers using the excess utility to make a solution that is a Pareto improvement

	Optimal Allocation	2019 Allocation
Total Utility	176,754.2	102,474.4
Efficiency Gap	-13,712.2	-
Coefficient of Variation	1.52	0.67
Gini Coefficient	0.65	0.37

Table: Comparison of Optimal and 2019 Allocations

Winter Routing Model

- Over winter, we see that the constrained allocation has the highest utility as expected since the model will focus on serving high demand sites first
- However, the constrained allocation performs even worse on the fairness metrics
  - A Gini coefficient of 0.8 is above even the most unequal societies on Earth
  - The efficiency gap is almost as large as the full-year allocation previously
- Despite the highly unequal allocation, it is again possible to construct transfers to remedy these issues using the generated surplus

	Constrained Allocation	Unconstrained Allocation	2019 Allocation
Total Utility	51259.8	43959.2	25224.47
Efficiency Gap	-12704.86	-3326.38	-
Coefficient of Variation	2.10	1.50	0.67
Gini Coefficient	0.80	0.64	0.37

Table: Comparison of Constrained, Unconstrained, and 2019 Allocations

Fisher Market

- The stochastic allocation program performed pretty poorly here, with a gap ratio of 93.3%
- This can largely be explained by the large standard deviations in arrivals
- Coefficients of variation for arrival types range from 1.95 to 5.55, which indicate extremely variable distributions

	Hindsight Solution	Myopic Solution
Total Utility	5100.74	341.9
Hindsight Gap	-	4758.8
Hindsight Gap Ratio	-	93.3%

Table: Comparison of Hindsight and Myopic Solutions

Fisher Market

- The myopic solution performed moderately well with a total maximum envy of 85.60297
- There is significant variation across types however
  - Types 2xx (types with the highest income) have 0 envy since they receive the lowest allocation everywhere
  - Types 0xx (types with the lowest income) have the greatest envy due to variation in allocation
- The hindsight solution has no envy since it allocations equally across all sites

Table: Maximum Envy by type for the Myopic Solution

Type	**Maximum Envy**
220	0.00000000
021	23.21401078
022	22.34700982
020	23.99665881
122	4.78431163
221	0.00000000
120	5.75402328
121	5.50696099
222	0.00000000

Implications

- Overall, a clear efficiency-fairness tradeoff emerged
- Models that increased aggregate utility always did so at the expense of other sites
- This ultimately matters for determining the purpose of the food bank itself
    - Is the goal to serve the most people possible?
    - Or is the goal to ensure as broad coverage as possible?

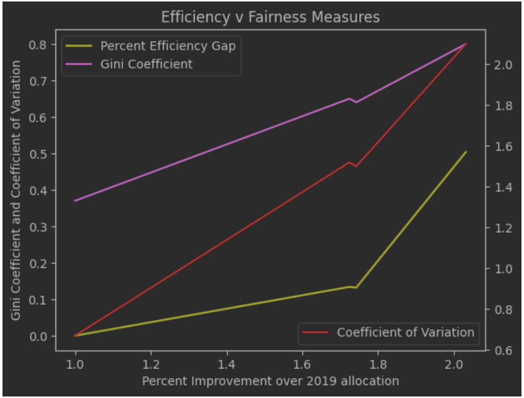

Figure: Efficiency versus Fairness metrics